SnowEcho: End-to-End Blind Skiing Assistance System Using Spatial Audio for Simulated Dynamic Slope

Anonymous Author(s)

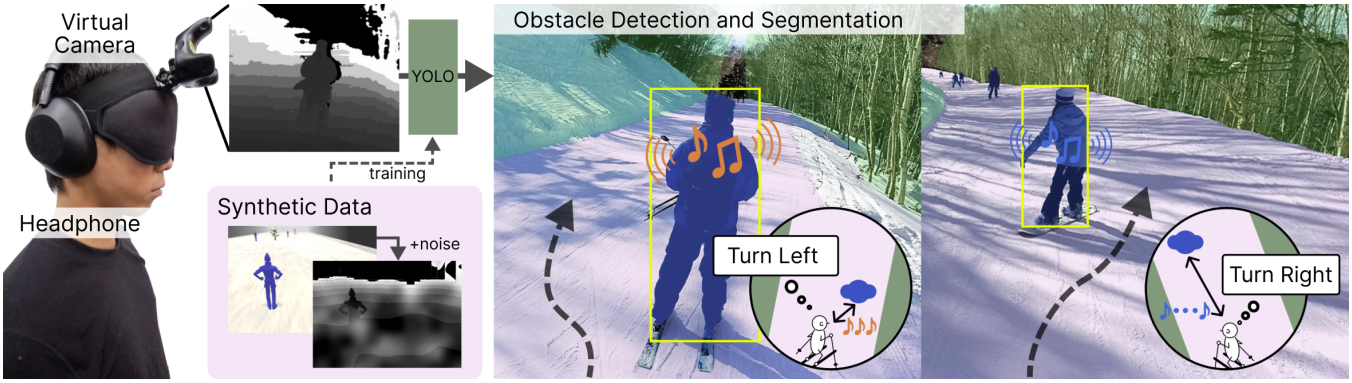


Figure 1: Overview of SnowEcho, an end-to-end blind skiing assistance system for simulated skiing. A head-mounted virtual camera captures the skier's viewpoint, and depth images augmented with real-slope-inspired noise are used for obstacle detection and segmentation. The system then renders spatial audio cues that convey obstacle direction and distance, while side cues indicate course boundaries. Rather than prescribing a single avoidance route, SnowEcho externalizes surrounding hazards so that skiers can perceive traversable space and choose their own line.

Abstract

Blind skiing usually relies on a sighted guide, limiting opportunities for independent participation. Prior ski assistance systems mainly use route guidance and assume static obstacles or ideal sensing, making them less suitable for crowded, dynamic slopes. We present **SnowEcho**, a skiing assistance system for people with visual impairments (PVI) that uses spatial audio to convey obstacle direction and distance, allowing skiers to choose their own line. In a simulator study with visually impaired skiers, SnowEcho improved obstacle avoidance and increased clearance in dense dynamic environments compared with a route-guidance baseline, while also improving perceived independence and consistency. We further tested an end-to-end sensing pipeline with depth-based detection and segmentation. Performance remained close to oracle conditions under moderate noise, but severe noise reduced clearance, confidence, and feedback consistency. These findings indicate both the promise of spatial audio scene awareness and the importance of cue stability for real-world deployment.

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CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools**.

Keywords

blind skiing, visual impairment, ski simulator, spatial audio, real-time sensing

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1 Introduction

People with visual impairments (PVI) participate in a wide range of sports, but access to sports remains constrained when activities rely heavily on visual information [5, 34, 52]. Skiing is one such example. In blind skiing, a sighted guide skis close to the blind skier and provides simple verbal cues such as "right", "left", and "keep going" [17]. Still, opportunities for PVI to ski remain limited due to the unavailability of experienced sighted guides.

Our ultimate goal is to realize a wearable system that enables PVI to ski independently on real slopes. Recent research has explored ski exergames and simulator-based training systems for visually impaired users [19, 36, 39]. However, they still do not fully reproduce the conditions required for real-world independent skiing.

One limitation of conventional route guidance ski assistance systems for the visually impaired [19, 36] is that they are not adapted to dynamic obstacles such as snowboarders and skiers that may

be present in real-world environments. When there are dynamic obstacles, the planned route may need to be updated repeatedly, making the feedback unstable and difficult to follow [19]. By contrast, systems that convey surrounding obstacle information and let users choose how to move [13, 46, 54, 55] remain underexplored in blind skiing.

Another limitation is the gap between *simulator assumptions* and *real-world deployment*. Prior blind skiing systems have mainly been designed and evaluated in simulator settings or under idealized sensing assumptions [19, 36]. Yet, a wearable system that operates on an actual ski slope must tolerate real sensing constraints, including limited field of view, viewpoint changes caused by head motion, finite sensing range, reduced spatial resolution, unstable boundaries, and missing depth values [19, 33].

To address these gaps, we present **SnowEcho** (Figure 1), an end-to-end blind skiing assistance system that shifts from route guidance to *spatial awareness*. Rather than telling users exactly where to go, SnowEcho conveys obstacles and course boundaries so that users can interpret the scene and choose their own actions. To better approximate real-world deployment, our system incorporates head-tracked viewpoints, depth noise informed by real ski-slope data, real-time detection and segmentation, and on-the-fly feedback generation.

We raise two research questions: **RQ1:** Can spatial audio feedback support more effective obstacle avoidance and a stronger sense of agency than conventional route-guidance approaches in dynamic environments? **RQ2:** When the system is driven by an end-to-end sensing pipeline rather than Oracle Value, how much sensing noise can be tolerated before skiing performance degrades?

To answer these questions, we conducted two user studies. First, we compared SnowEcho with a route-guidance baseline in a ski simulator with visually impaired skiers. Second, we evaluated the system using an end-to-end sensing pipeline under multiple levels of sensing noise that approximate real deployment conditions.

Our results show that, in dense dynamic environments, SnowEcho improves obstacle avoidance and increases the minimum distance maintained from obstacles relative to a route-guidance baseline. The results also suggest that the proposed method supports a greater sense of independent skiing and agency. Furthermore, when feedback is driven by noisy sensing rather than oracle scene information, performance remains close to *Oracle Value* under *Moderate Noise*, whereas *Severe Noise* reduces clearance, confidence, and perceived consistency. Together, these findings suggest that spatial-audio-based scene awareness could be one promising direction for enabling safer and more autonomous blind skiing.

2 Related Work

2.1 Sports Assistance for People with Visual Impairments

To broaden the range of sports accessible to PVI, prior work has explored activities such as yoga [50], badminton [27], and exergames [12, 37, 38]. In addition, to enable PVI to participate in sports without relying on sighted guides, systems have been developed for activities including climbing [23, 47, 51], running [11, 16, 49, 56], and swimming [40, 44].

Table 1: Comparison of the prior skiing system and our system. Our approach enables independent skiing by presenting dynamic obstacles using spatial audio. "Oracle": systems that use Oracle detection; "Assistant": systems that give feedback based on the input of sighted guides.

System	Independent Use by PVI	Feedback Modality	Feedback Information	Obstacle	Input
S-SH [43]	✗	LED	**Obstacle**	**Dynamic**	LiDAR
Aggravi *et al.* [1]	✗	Haptic	Route	**Dynamic**	Assistant
VIHapp [17]	✗	Haptic	Route	**Dynamic**	Assistant
Miura *et al.* [36]	✓	Stereo Audio	Route	Static	Oracle
SlopeNav [19]	✓	Stereo Audio	Route	Static	RGB
SnowEcho	✓	**Spatial Audio**	**Obstacle**	**Dynamic**	**Depth**

In the context of skiing, haptic feedback has been explored to facilitate communication between a guide and a skier with visual impairments [1, 17]. However, these approaches primarily aim to reduce the burden on sighted guides. To enable independent skiing without a guide, prior work has explored ski exergames [39] and simulator-based training systems [19, 36]. Table 1 summarizes prior skiing assistance systems and positions SnowEcho relative to them. For instance, SlopeNav [19] introduced a system that detects obstacles and provides navigation assistance within a simulator. Nevertheless, these systems are designed under the assumption of fixed environments or static obstacles, leaving a significant gap between simulated and real-world conditions.

2.2 Navigation for Obstacle Avoidance

Navigation methods that convey obstacle avoidance paths to PVI can be broadly categorized into two types: route guidance and obstacle avoidance. In route guidance approaches, the system plans a specific avoidance path and directly instructs the user where to move. In contrast, obstacle avoidance approaches provide information about the surrounding environment, allowing the user to decide how to navigate.

For route guidance, prior work has proposed systems that present directional guidance using haptic or auditory cues in walking [7, 26, 29], driving [45] and running [11, 16, 32, 56], as well as in skiing [19, 36]. In these approaches, users can simply follow instructions, resulting in high efficiency and low cognitive load [48]. However, such methods limit users' ability to explore the environment and make independent decisions in recreational contexts such as skiing. Moreover, automatically generating appropriate routes becomes challenging in complex environments, particularly those involving dynamic obstacles.

Alternatively, obstacle avoidance approaches support movement by mapping sensed environmental information to auditory or haptic feedback. Presti *et al.* [46] proposed a system that assists visually impaired pedestrians by conveying the distance and direction of obstacles in real-time. Similarly, in racing games [13, 54] and virtual environments [42], sound-based representations of the environment enable users to actively explore and choose their paths. However, the use of such approaches in skiing remains limited. For sighted users, systems have been proposed to warn of approaching obstacles from behind [43] and to present map-based information [10, 21]. However, these systems rely on visual feedback and have not been adapted for users with visual impairments.

2.3 Auditory Feedback for Spatial Perception

One common auditory feedback is the use of speech-based feedback. Prior work has proposed systems that convey information such as the position and orientation of surrounding objects [6, 20, 24, 41] and navigation instructions [7, 9, 26, 29, 31] through synthesized speech. While speech-based feedback can convey complex information, it has the drawback of requiring time to deliver the message.

Another approach represents spatial information through sonification. Compared to speech, non-verbal sounds are more robust to background noise, impose lower cognitive load, and are independent of language [8, 28]. Prior studies have mapped environmental information directly to sound, including mapping images to sound [35], echolocation [22], and spatial audio [14]. In addition, some approaches encode specific properties of obstacles into sound to facilitate learning [2, 13, 15, 46, 53–55]. However, compared to walking or indoor navigation, skiing introduces additional challenges due to higher speeds and longer interaction distances.

Building on these approaches, we design auditory feedback tailored for skiing scenarios. Our system detects obstacles and presents their direction using spatial audio, while conveying distance through variations in frequency and pitch.

3 SnowEcho

3.1 System Direction Derived from Pilot Study and Existing Research

Our system is designed to support continuous spatial awareness through **spatial audio** rather than explicit route prescription. Prior approaches often emphasized route guidance, whereas our goal is to externalize the geometry of the scene without prescribing a single action. To design the auditory feedback, we conducted a pilot study with six sighted participants wearing blindfolds. The results revealed two key findings:

- When multiple sounds of the same type are heard at once, users may be unable to process them adequately, which could result in confusion.
- Since obstacles overtaking from behind are avoided automatically, simpler sounds are sufficient compared with those used for obstacles in front.

The spatial feedback is designed to provide continuous environmental awareness without overloading the skier's attention. Front obstacles are rendered as spatialized sources whose urgency changes with distance, course boundaries are conveyed as continuous lateral cues, and rear overtaking events are reduced to a brief side-specific alert. Rather than monopolizing the skier's attention with symbolic instructions, the system maintains a lightweight but persistent representation of front hazards, side limits, and rear passing. As a result, the skier can interpret the scene continuously and decide how to turn, preserving the sense of agency that is central to enjoyable and independent skiing.

To enable blind skiing in real-world environments, we developed a depth informed sensing pipeline tailored to outdoor snow scenes. As prior work on blind ski assistance has mainly focused on simulator settings [19, 36], real-world sensing pipelines remain less explored. In this work, we capture visual data in an actual ski resort using a stereo depth camera mounted on the head and analyze the dominant noise in outdoor snow scenes. We then reproduce these noise patterns in a ski simulator before inference and evaluate their impact on the system safely in a controlled environment.

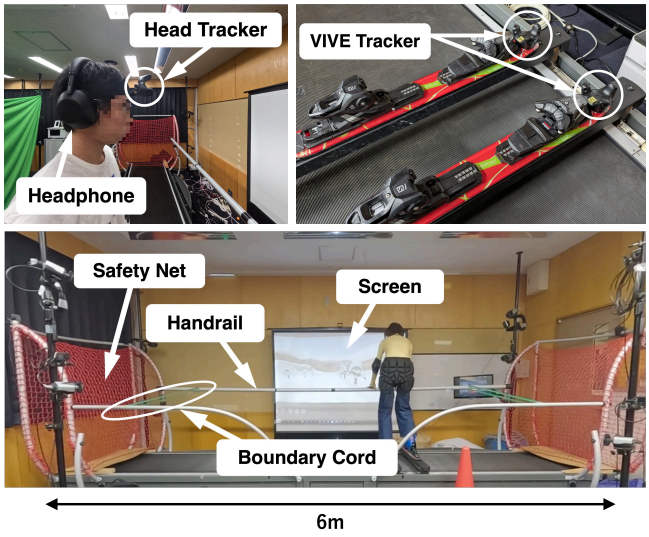


Figure 2: Hardware configuration of the simulator-based evaluation. VIVE trackers on the skis and head capture user motion and viewpoint, while headphones present spatial audio cues. The full setup includes a projected course display, handrail, safety nets, and vinyl boundary cords, enabling safe evaluation in a controlled skiing environment.

3.2 Hardware

To approximate a realistic skiing environment, we used an indoor ski simulator (SkyTechSport Ski simulators)[1]. The simulator is 6 m wide and uses equipment comparable to that used in actual skiing. For safety, a handrail is installed in front of the simulator, and safety nets are installed on both sides. Participants were allowed to use the handrail during the experiment. Vinyl cords were attached to the inner side of the nets as a boundary indicating the user's proximity to the edges before contact (Figure 2).

The skis are mounted in parallel with a fixed separation. When the participant tilts the skis, the skis are pulled by wires from the tilted direction. We attached VIVE trackers to the tips of both skis to track the participant's position. For auditory feedback, we used headphones (SONY WH-1000XM6) to reduce latency and improve spatial audio perceptibility. During the user studies, a screen was placed in front of the participant to project the course scene.

3.3 Auditory Feedback

As shown in Figure 3, to convey positions of both static and dynamic obstacles, we prepared three types of sounds:

Object Audio: This feedback presents the locations of obstacles within 8 m in front of the user with a simple sinusoidal sound. Let $\mathbf{p}_{o,i}$ and $\mathbf{p}_u$ denote the positions of obstacle i and the user,

[1] https://skytechsport.com/ski-simulators-home/

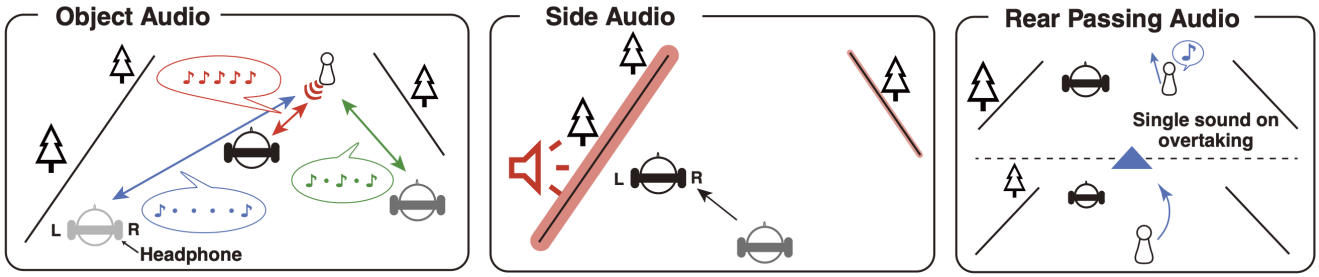


Figure 3: Three auditory cues used in SnowEcho. Object Audio spatially indicates front-obstacle direction and increasing urgency with decreasing distance; Side Audio continuously indicates the nearest course boundary; Rear Passing Audio briefly indicates the side of an overtaking skier. This design supports scene awareness without explicit turn-by-turn route guidance.

respectively, and let $\mathbf{f}$ denote the user's forward direction. We provide this feedback only for obstacles that satisfy

$$d_i = \|\mathbf{p}_{o,i} - \mathbf{p}_u\|_2 \le 8, \qquad (\mathbf{p}_{o,i} - \mathbf{p}_u) \cdot \mathbf{f} > 0. \tag{1}$$

The obstacle direction is conveyed via spatial audio using a head-related transfer function (HRTF), while the distance is represented by the sound's emission frequency and volume. The resulting binaural cue can be written as

$$y_{L/R,i}(t) = V_{\text{obj}}(d_i)\,\big(x_i * h_{L/R}(\phi_i, \theta_i)\big)(t), \tag{2}$$

where $x_i(t)$ is the sinusoidal cue, $h_{L/R}$ is the left/right HRTF, and (ϕ_i, θ_i) denote the obstacle direction relative to the user. The emission frequency of the sound changes at distances of 8, 6, 4, and 2 m. As the user approaches an obstacle, the sound is emitted more frequently and at a higher volume to indicate increasing risk. When the obstacle enters within 2 m, the pitch also increases to emphasize the imminence of a collision.

Side Audio: To indicate the edge of the course, a continuous sinusoidal sound is emitted from the direction of the nearest side of the slope within 5 m. Let x_p denote the user's lateral position and let x_L and x_R denote the left and right boundaries of the slope. The distance to the nearest side is

$$d_{\text{side}} = \min\big(|x_p - x_L|,\ |x_R - x_p|\big). \tag{3}$$

The cue is emitted only when $d_{\text{side}} \le 5$, and its amplitude increases as the user approaches the side. The sound pitch is designed to be different from that of Object Audio to make a clear distinction.

Rear Passing Audio: We prepared an additional sound to notify participants of obstacles overtaking them from behind. This sound was played when the obstacle passed the user, and it indicated the overtaking side (e.g., if the obstacle passed on the right, the sound was presented through the right headphone).

In addition to Object Audio, Side Audio and Rear Passing Audio, we implemented three other types of auditory feedback: (1) a notification sound upon collision with an obstacle, (2) a countdown sound indicating the start of the system, and (3) an end sound indicating the termination of the system. These sounds were also introduced in the prior work SlopeNav [19], and we adopted them in this study to clearly convey operational boundaries and changes in system state.



Figure 4: Images captured on a real ski slope using a head-mounted OAK-D camera: an RGB image from the skier's viewpoint (left) and a depth image showing characteristic sensing artifacts in snow scenes (right).

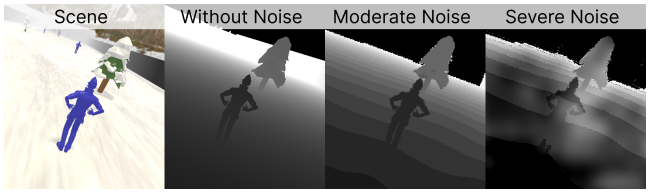


Figure 5: Scene from the synthetic dataset and results of Depth Augmentation. The image is tilted because the system uses head-tracked viewpoints.

3.4 Bridging the Data Domain Gap

To approximate real-world sensing within the simulator, we constructed a pipeline with two stages: head-tracked viewpoint control and depth augmentation derived from real ski-slope recordings. We first collected data in an actual skiing environment using an OAK-D[2] stereo camera mounted on the head and analyzed the dominant noise characteristics introduced in the real world (Figure 4). We then reproduced these characteristics as depth augmentation, generating synthetic training data (Figure 5).

3.4.1 Head-tracked Viewpoint Control. To approximate the skier's field of view, we attached a VIVE Tracker to the user's head and used it to control the orientation of the reference camera in Unity. We

[2]https://docs.luxonis.com/hardware/products/OAK-D

set the reference-camera field of view to 70° to match the viewing angle of the head-mounted OAK-D camera used in the real-world data collection.

3.4.2 Depth Augmentation. We analyzed the depth videos captured on real ski slopes with the head-mounted OAK-D camera (Figure 4) to identify sensing artifacts that are likely to affect skiing assistance. Compared with ideal simulator depth, the recorded depth images showed several recurring characteristics. First, fine geometric structure was often smoothed, suggesting limited effective spatial resolution. Second, the snow surface frequently appeared in piecewise-constant layers rather than as a perfectly continuous surface. Third, depth values were particularly unstable around object contours and occlusion boundaries, where foreground and background depths were often mixed. Fourth, the reliability of depth estimates decreased with distance, making farther obstacles less precisely localized. Finally, invalid or truncated measurements appeared in out-of-range regions and other low-confidence areas. These characteristics are important in our setting because they directly affect obstacle localization and the temporal stability of the spatial audio feedback. Based on this analysis, we reproduced these characteristics in the simulator as follows.

We created a simulation scene in Unity and computed metric depth from it. The depth map was downsampled to 320×200 and then resampled to the original resolution, producing blocky local structures and mild spatial smoothing that approximate the limited spatial resolution of recorded depth images. We then apply additional corruptions before exporting the depth image:

Depth Quantization: To reproduce the layered appearance observed on the snow surface, we discretize continuous depth values into coarse depth layers. As a result, smoothly varying surfaces become stepped, which reflects the piecewise-constant appearance often observed in the recorded depth images. In our current configuration, we quantize depth with a 0.5 m step.

Boundary Instability: To reproduce the instability observed around depth discontinuities, we perturb pixels near strong depth edges, such as object contours and occlusion boundaries. This models the fact that depth estimates are typically least stable around edges, where mixed pixels and partial occlusions are common. In our implementation, boundary noise is generated either by borrowing a neighboring depth value across the discontinuity or by applying local random jitter.

Distance-dependent Noise: To reproduce the increased uncertainty in farther regions, we add noise whose magnitude increases with distance from the sensor. This reflects the reduced reliability of distant depth measurements and the resulting uncertainty in reconstructed cue positions for far obstacles.

Missing and Truncated Depth Values: To reproduce invalid or out-of-range measurements observed in real depth sensing, we allow such measurements to appear as missing depth values. This reproduces dropout-like failures and the finite sensing range of real-world depth cameras. In this research, depth is cropped beyond 10 m for the inference image.

3.4.3 Depth Augmentation Evaluation. To assess how closely the augmented images approximate the real-world distribution, we generated 500 images for each of three noise conditions (*Without Noise*, *Moderate Noise*, and *Severe Noise*) and compared them with 500 real images captured in the actual environment. These noise conditions are identical to those used in Study 2 (Section 5). Qualitative examples are shown in Figure 5.

We used $\mathrm{FID}_{\mathrm{CLIP}}$ [30] as the evaluation metric. The results show that *Without Noise* yields 32.143, *Moderate Noise* 25.694, and *Severe Noise* 22.689. These findings indicate that applying noise shifts the image distribution closer to that of real-world data.

3.5 Obstacle Sensing on Depth Image

Using both the synthesized and real-world datasets, we developed an obstacle detection model and integrated it with the feedback system.

3.5.1 Detector Training. We employed YOLOv8n-seg [25] as the obstacle detection model. First, we constructed a synthetic dataset of 5,000 images featuring obstacles such as people and trees, using the pipeline described in Section 3.4 (3,000 training, 1,000 validation, and 1,000 test images).

For fine-tuning and evaluation, we developed a real-world dataset of 750 annotated images (450 training, 150 validation, and 150 test images). After fine-tuning, the model achieved an mAP50 of 0.895 and a mean distance error of 0.939 m. This performance is comparable to the *Moderate Noise* condition in the synthetic dataset, supporting the validity of our noise-level modeling in Study 2 (Section 5).

3.5.2 Detection and Feedback Integration. During runtime, the system uses only the depth image captured from the head-tracked viewpoint. The depth image is sent from Unity to a local detection server, where YOLOv8n-seg estimates obstacle instances together with their image positions and depths. The server returns, for each detection, the object class, confidence, image center, and estimated distance.

The server decodes the 8-bit depth PNG into metric forward depth by

$$D(u,v) = \frac{d_{\max}}{255}\,\tilde{I}(u,v), \tag{4}$$

where $\tilde{I}(u,v) = I(u,v)$ (or $255 - I(u,v)$ for inverted encoding), and $d_{\max}$ is the shared decoding upper bound.

For each detected instance i with mask $\mathcal{M}_i$, valid in-mask depths are

$$\mathcal{V}_i = \{D(u,v) \mid (u,v) \in \mathcal{M}_i,\ D(u,v) > 0\}, \tag{5}$$

and the representative depth is

$$d_i = P_{20}(\mathcal{V}_i). \tag{6}$$

We selected P_{20} after trying several percentile values in preliminary tests, as it appeared to offer a reasonable trade-off for our setting. Very low percentiles were more susceptible to sporadic under-estimated pixels, while higher percentiles were more influenced by boundary-mixed background depths. In practice, P_{20} provided a depth estimate close to the nearest visible obstacle surface with sufficient temporal stability for collision reasoning.

Unity then reconstructs a 3D cue position from the returned image coordinates and depth. Let (u_i, v_i) denote the normalized image coordinates of detection i, let $\hat{\mathbf{r}}_i$ denote the camera ray passing through that pixel, and let d_i denote the estimated forward depth. The corresponding 3D cue position is computed as

$$\mathbf{p}_i^{\text{cue}} = \mathbf{o}_c + \frac{d_i}{\hat{\mathbf{r}}_i \cdot \mathbf{f}_c} \hat{\mathbf{r}}_i, \tag{7}$$

where $\mathbf{o}_c$ and $\mathbf{f}_c$ are the camera origin and forward direction. This position is then used to generate spatial audio feedback in Unity. Obstacle detections drive Object Audio, while wall detections drive Side Audio.

Because depth estimates can fluctuate near the alert thresholds, directly mapping each frame to an alert level can produce unstable feedback. To improve robustness, we update the ObjectAlert level only after the same candidate level is observed for multiple consecutive frames. This reduces rapid switching caused by depth noise while preserving responsiveness.

We measured end-to-end latency from depth-image acquisition to audio feedback during the sensing-based conditions in Study 2 (Section 5). Under *Moderate Noise*, the mean latency was 67.44 ms (SD = 11.39), and under *Severe Noise*, the mean latency was 70.76 ms (SD = 12.65).

4 Study 1: Verification of the Effectiveness of our System with Spatial Audio

Study 1 evaluated the interaction design under an oracle-scene-state setup. For both SnowEcho and SlopeNav, obstacle and boundary locations were obtained directly from the simulator state in Unity. To verify whether spatial audio is effective in environments with dynamic obstacles, we conducted a user study with 11 participants with visual impairments who had skiing experience, as summarized in Table 2 (age: 22–63 years, mean = 46.1; 8 males and 3 females). Because participation required both having a visual impairment and routinely engaging in skiing, we recruited participants with the cooperation of two blind clubs. Each study lasted approximately 2.5 hours, and participants received an honorarium of approximately USD 56, paid in local currency. This study was conducted with approval from the institutional research ethics committee.

4.1 Task and Conditions

Participants were asked to complete a task on a ski simulator in which they advanced along a course while avoiding obstacles and performing turns by shifting their body left and right. Because preserving even limited residual visual information can provide participants with low vision with an experience closer to real-world skiing, the course scene was projected onto a screen placed in front of each participant. We compared two feedback strategies. SnowEcho, our spatial-awareness method, and SlopeNav [19], a route-guidance baseline. SnowEcho presented obstacle direction through spatialized Object Audio and obstacle distance through temporal and intensity changes. It also conveyed the nearest course boundary using Side Audio and informed users of rear overtaking events using Rear Passing Audio. SlopeNav generated an avoidance route and instructed the user where to move via auditory cues: right-ear audio for right turns, left-ear audio for left turns, and binaural audio for going straight, with pitch indicating turn magnitude. We prepared three courses:

(1) C_{Sparse}: A course with 10 dynamic obstacles randomly placed at 15±3 m intervals. Obstacles consisted of two skier types: zigzagging and straight-line skiers.
(2) C_{Dense}: A course with 20 dynamic obstacles randomly placed at 8±3 m intervals.
(3) C_{Real}: A course with 20 static and dynamic obstacles randomly placed at 10±2 m intervals. We also introduced skiers who occasionally overtook the participant from behind, passing on either side and automatically avoiding collisions. These were included to elicit sudden feedback and create more challenging, realistic conditions.

Table 2: Participant demographics. In the Gender column, M denotes male, and F denotes female. In the Eyesight column, LV denotes low vision. In the Skiing Level column, Intermediate refers to participants who can perform parallel or wedge turns, whereas Expert refers to participants who can ski on any course.

ID	Age	Gender	Eyesight	Age of Onset	Skiing Exp. (years)	Skiing Level	Simulator Exp. (times)
P01	22	F	LV	10	15	Intermediate	0
P02	44	M	LV	15	12	Intermediate	1
P03	38	M	LV	5	5	Expert	0
P04	56	M	Blind	13	43	Expert	2
P05	60	M	Blind	16	35	Intermediate	0
P06	51	F	Blind	4	3	Intermediate	0
P07	55	F	Blind	0	30	Intermediate	0
P08	63	M	Blind	22	60	Expert	2
P09	25	M	Blind	0	5	Intermediate	0
P10	35	M	Blind	1	20	Expert	0
P11	58	M	LV	40	40	Expert	0

4.2 Procedure

We first conducted a pre-study interview with each participant to collect their demographics and skiing experience. We then introduced the ski simulator to the participants and they used it for approximately 10 minutes to familiarize themselves with the system. We explained the feedback details of the first system and then allowed participants to practice on each of the three prepared courses. To counterbalance order effects, five participants used SnowEcho first, while six used SlopeNav first. Participants then completed one run on each course, and we evaluated their performance. The order of C_{Sparse}, C_{Dense}, and C_{Real} was randomized, and participants were not informed which course they were skiing during the trials. Participants completed questionnaires about the system they experienced, including NASA-TLX and SUS. Then, they conducted the task with another system. After completing all trials, participants answered an overall post-study questionnaire and took part in a semi-structured interview.

During the trials, participants were allowed to use the handrail installed on the ski simulator. However, to prevent them from estimating their position by continuously holding the same location, we instructed them to adjust their grip position on the handrail as they moved.

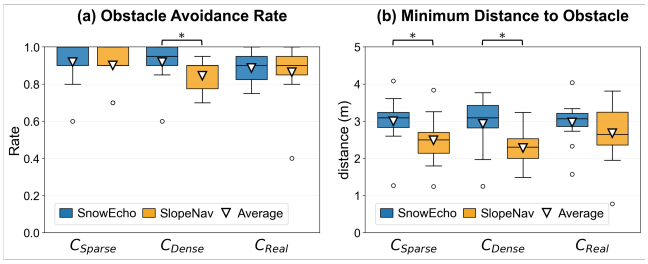


Figure 6: Results of Study 1 showed that SnowEcho reduces collisions and improves clearance from obstacles. Box plot of obstacle avoidance rate and mean minimum distance to obstacles in each experimental condition. * indicates a significant difference ($p < .05$).

4.3 Metrics

In this experiment, we designed our evaluation metrics by referring to those used in the SlopeNav study [19].

Obstacle Avoidance Rate: This metric evaluates the efficiency of obstacle avoidance and is defined as the percentage of obstacles that the user successfully avoided.

Minimum Distance to Obstacle: This metric measures the minimum distance maintained from each obstacle when the participant passed it. We then averaged these minimum distances for each course. Since greater separation from obstacles is generally considered safer for skiers with visual impairments, larger values indicate better performance for this metric.

Quantitative Questionnaire: After each condition, participants completed the System Usability Scale (SUS) [3] and the NASA Task Load Index (NASA-TLX) [18]. The SUS uses a 5-point Likert scale for usability, and the NASA-TLX uses a 10-point scale for perceived workload. Participants also rated confidence, sense of agency, and feedback consistency on 7-point Likert scales.

Qualitative Evaluation: Through the interviews, we elicited participants' perceptions of each system, including its advantages, drawbacks, and areas for improvement.

4.4 Results

4.4.1 Quantitative Evaluation. Figure 6 shows the obstacle avoidance rate and mean minimum distance to obstacles for each course. We compared SnowEcho and SlopeNav using the Wilcoxon signed-rank test.

For the obstacle avoidance rate, SnowEcho achieved an obstacle avoidance rate of approximately 90% across all three courses. In C_{Dense}, the proposed method significantly outperformed SlopeNav ($p < .05$). For the minimum distance, SnowEcho maintained a significantly greater distance than SlopeNav in C_{Sparse} and C_{Dense} ($p < .05$). No significant differences were found in C_{Real} for either metric.

For NASA-TLX, SnowEcho scored 5.03 ± 2.29, lower than SlopeNav (5.55 ± 1.77), although the difference was not statistically significant. For SUS, SnowEcho scored 67.3 ± 20.0, higher than SlopeNav (61.4 ± 18.8), also without a significant difference.

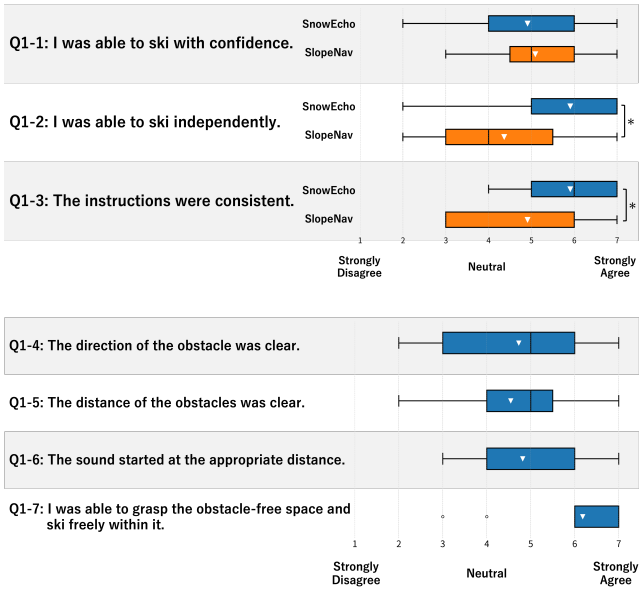


Figure 7: Results of Study 1 showed that SnowEcho enhanced users' sense of agency and provided more consistent guidance. Results of the Likert-scale questionnaire (1 = strongly disagree, 7 = strongly agree). * indicates a significant difference ($p < .05$).

Figure 7 (Q1-1~Q1-3) shows participants' ratings on confidence, independence, and consistency. SnowEcho received higher ratings than SlopeNav across all items. In particular, independence and consistency showed clear improvements, with distributions shifted toward stronger agreement and statistically significant differences. Figure 7 (Q1-4~Q1-7) presents results for the proposed method. Participants reported especially high ratings for perceiving obstacle-free space, with a mean score above 6.

4.4.2 Qualitative Evaluation. Six participants reported that SlopeNav, which provides only travel direction, made it hard to notice obstacles, whereas SnowEcho improved obstacle awareness. Several participants also noted that left–right sound localization became clearer as an obstacle approached: **C1**: *"At the moment the sound starts, it is difficult to judge whether the obstacle is on the left or right, but as it gets closer, the direction becomes clearer."* **(P08)**

Regarding the audio mapping for distance/risk, several participants said it was not immediately intuitive and would require getting used to. Five participants evaluated Side Audio as highly effective, saying it helped identify whether an obstacle was coming from the left or right and made the distance relationship easier to understand: **C2**: *"It was helpful because I could tell from the sound whether an obstacle was coming from the left or right, and it was also easier to understand the distance relationship."* **(P01)**

In contrast, eight of the eleven participants said that feedback for obstacles overtaking from behind was unnecessary. They explained that they could not fully process the additional sound, or that they were already concentrating on obstacles in front and

therefore often did not clearly hear the rear-passing alert. Several participants also noted that skiers approaching from behind would usually avoid them automatically, reducing the need for explicit feedback. Some added that although snowboarders coming from behind felt particularly frightening, they could usually notice them even without system feedback because the sound they produced was already loud.

Overall, these comments suggest that some participants preferred SnowEcho not only for obstacle avoidance but also because Side Audio helped them maintain awareness of the course boundaries and perceive traversable free space, allowing them to choose their own line more freely. By contrast, several participants preferred SlopeNav because its turn-by-turn guidance was closer to their usual skiing style.

5 Study 2: Evaluation with a More Realistic Sensing Pipeline

Study 2 tested whether SnowEcho still works when the feedback comes from the sensing pipeline rather than from *Oracle Value*. We compared one oracle condition with two noisy-sensing conditions.

Five participants from Study 1 took part in this follow-up study (P01, P06, P07, P08, and P10). These participants were selected from the original cohort to examine the effect of sensing uncertainty while minimizing the confounding effect of learning a new interaction technique. The five participants included both blind and low-vision skiers with prior skiing experience and prior experience with our simulator setup. Each study lasted approximately 2 hours, and participants received an honorarium of approximately USD 38, paid in local currency.

5.1 Conditions

We prepared three within-subject conditions: *Oracle Value*, *Moderate Noise*, and *Severe Noise*.

In the *Oracle Value* condition, auditory feedback was generated directly from the simulator state in Unity, identical to the configuration used in Study 1. That is, obstacle and boundary locations were obtained directly from the simulation, without invoking the sensing, detection, or segmentation pipeline. We use this condition as a reference condition for the idealized case in which the system has direct access to the scene state. *Moderate Noise* and *Severe Noise* used the sensing pipeline in Section 3.5. They differed only in the amount of depth noise added before detection.

5.1.1 Moderate Noise. We added mild depth errors before detection. Depth values were rounded in 0.5 m steps. Depth near object edges was slightly unstable. We also added depth noise that became larger at longer distances, up to about 0.3 m at 10 m.

5.1.2 Severe Noise. We added stronger versions of the same depth errors. Depth values were rounded in 1 m, edge noise was larger, and distance-based noise reached up to about 4.0 m at 10 m.

5.2 Task

Participants performed the same skiing task as in Study 1. To control for course difficulty, we used a single course based on C_{Real}, containing 20 static and dynamic obstacles placed at intervals of 10 ± 2 m. Rear-overtaking obstacles were removed based on the feedback from Study 1, allowing the evaluation to focus on obstacles in front of the participant and on course boundaries. Obstacle layouts varied across runs, but overall difficulty was kept constant.

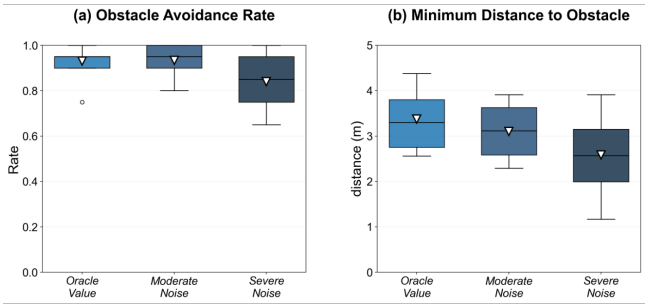


Figure 8: Results of Study 2 on obstacle avoidance rate and mean minimum distance across conditions. Performance remains close to *Oracle Value* under *Moderate Noise*, while *Severe Noise* reduces obstacle avoidance and clearance.

5.3 Procedure

Because all participants had already completed Study 1, we first provided a brief refresher on the ski simulator and the auditory feedback. Participants then practiced basic lateral movement on the simulator and completed a short training phase to reacquaint themselves with the system. This training included a boundary-only practice and a short practice run in the sensing-based setup.

After training, participants completed the task under all three conditions. For each condition, they performed three runs, resulting in a total of nine runs per participant. Obstacle layouts differed across runs, while overall course difficulty was kept constant. The order of the three conditions was counterbalanced across participants. After each condition, participants completed a post-condition questionnaire and a short break.

At the end of the study, we conducted a semi-structured interview about the differences they perceived across the three conditions and about their experience using the sensing-based system.

5.4 Metrics

For quantitative evaluation, we used *Obstacle Avoidance Rate* and *Mean Minimum Distance to Obstacles*. To assess subjective workload, participants completed the Raw NASA-TLX [4], using a 10-point rating scale for each item. Furthermore, participants rated their experience on 7-point Likert scales regarding their confidence while skiing, sense of security, and the perceived consistency of the instructions. Finally, a semi-structured interview was conducted post-study to elicit qualitative insights.

5.5 Results

5.5.1 Quantitative Evaluation. Figure 8 summarizes the objective performance in the *Oracle Value*, *Moderate Noise*, and *Severe Noise* conditions. For obstacle avoidance rate, participants maintained high performance in both the *Oracle Value* and *Moderate Noise* conditions, exceeding 90%. In *Severe Noise*, the rate decreased to about

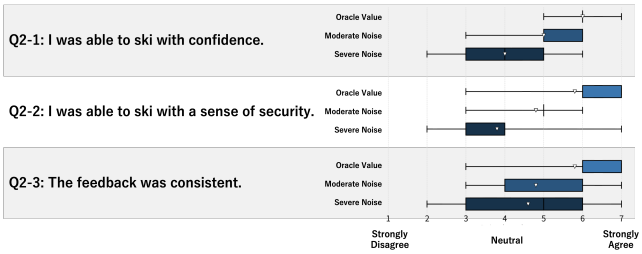


Figure 9: Results of Study 2 showed that increasing sensing Noise reduced users' confidence, security, and perceived feedback consistency. Box plots show Likert-scale ratings for Q2-1 to Q2-3 under three sensing conditions (1 = strongly disagree, 7 = strongly agree).

85%. This indicates that moderate sensing noise had little effect, while stronger noise reduced performance. A similar tendency was observed for the mean minimum distance to obstacles. As sensing noise increased, participants tended to pass closer to obstacles.

The Raw NASA-TLX ratings showed the same overall pattern. The mean of the six TLX items increased from 4.63 in *Oracle Value* to 4.93 in *Moderate Noise* and 6.00 in *Severe Noise*, suggesting that noisier sensing generally increased perceived workload.

Figure 9 summarizes participants' responses to three 7-point Likert-scale items assessing confidence, security, and perceived consistency of the feedback. Across all items, ratings gradually decreased from *Oracle Value* to *Moderate Noise* and further to *Severe Noise*. The decline was most pronounced for confidence, with security and perceived consistency showing similar downward trends.

5.5.2 Qualitative Evaluation. Interviews indicated that participants mainly differed in perceived feedback clarity and consistency across conditions. Four participants described *Oracle Value* as the clearest or easiest condition, while three identified *Severe Noise* as the most difficult. As P07 noted: **C3**: *"(In Severe Noise), what I perceived was clearly different from other conditions, and I did not know where to direct my attention. It lacked consistency."* **(P07)**

Across conditions, four participants reported relying mainly on left–right localization and adopting conservative strategies, such as staying near the center, making smaller turns, or continuing straight until the cue became clearer. P10 explained: **C4**: *"When I could not tell whether it was on the right or left, I kept going straight, and as it came closer it gradually became clearer; then I started to avoid it."* **(P10)**

Participants also reported that *Severe Noise* made some scene elements harder to interpret. Three participants said that dynamic obstacles were especially difficult because their motion was hard to predict, and at least three reported that traversable space was harder to understand, particularly near boundaries or between multiple obstacles. P08 contrasted the sensing-based conditions as follows: **C5**: *"(In Moderate Noise,) it was easy to understand the space between obstacles, and the open space was clear. (In Severe Noise), I sometimes collided before I could understand it."* **(P08)**

Taken together, these interviews suggest that our sensing pipeline preserved directional usefulness under *Moderate Noise*, while *Severe Noise* reduced cue consistency and made dynamic obstacles and free space harder to interpret. This pattern aligns with the quantitative results, where confidence declined and workload tended to increase as sensing noise became more severe.

6 Discussion

6.1 SnowEcho Supports Self-Directed Skiing Through Perceivable Free Space

Study 1 provides a clear answer to RQ1: in dense dynamic environments, spatial-audio-based scene awareness can support both more effective obstacle avoidance and a stronger sense of agency than conventional route guidance. SnowEcho improved avoidance performance and obstacle clearance in dynamic conditions, while also receiving higher ratings for independent skiing and consistency. These findings suggest that, for blind skiing, making free space perceptible may be more effective than prescribing a single route. A central implication of Study 1 is that the value of SnowEcho lies not only in improving obstacle avoidance, but in changing the form of assistance itself. This discussion is mainly grounded in the agency-related findings in Figure 7 and the qualitative results in Section 4.4.2, especially C2 and the subsequent interview summary. Rather than prescribing a single avoidance route, SnowEcho makes surrounding obstacles and course boundaries perceptible, allowing users to choose their own line. In this sense, the contribution of SnowEcho is not merely safer avoidance, but support for a more self-directed skiing experience through perceivable free space. This interpretation is also consistent with the performance trends in Figure 6, suggesting that this increase in agency does not simply come at the expense of safety.

6.2 Trade-offs Between Route Guidance and Spatial Awareness

Our comparison with SlopeNav suggests a key trade-off between direct guidance and user agency. SlopeNav tells users where to move, whereas SnowEcho presents surrounding obstacles and boundaries so that users can decide how to act. This difference appears especially important in skiing, where enjoyment and independence are closely tied to being able to choose one's own line. Consistent with this interpretation, SnowEcho improved performance in dense dynamic environments and received higher ratings for independent skiing. At the same time, several participants preferred SlopeNav because its turn-by-turn guidance was closer to their usual skiing style. These findings suggest that the advantage of SnowEcho is not only better obstacle avoidance in dynamic situations, but also its support for a more self-directed skiing experience, while route guidance may still suit users who prefer more explicit instruction.

6.3 Effectiveness of SnowEcho with Real-time Sensing

Study 2 answers RQ2 by showing that SnowEcho remains effective under *Moderate Noise*, but becomes challenging to use under *Severe Noise*. Under *Moderate Noise*, performance stayed close to *Oracle Value* condition: obstacle avoidance remained above 90%, and workload increased only slightly. This suggests that SnowEcho does not depend on perfect scene knowledge, as users can still rely on approximate but stable spatial cues to avoid obstacles safely.

One possible reason for this robustness is that our method conveys the surrounding scene as continuous spatial information, instead of prescribing a single avoidance route. In route-based guidance, even small sensing errors can lead to unstable or frequently changing instructions. In contrast, our approach can remain usable even when obstacle localization is somewhat inaccurate, as long as the general direction and urgency of the obstacle are preserved. This interpretation is also supported by the interview results: several participants reported that when the cue was initially ambiguous, they continued straight or made smaller turns until the left-right direction became clearer. These behaviors suggest that users could still exploit the feedback under moderate noise by adopting conservative but effective strategies.

At the same time, *Severe Noise* condition reveals an important limit of this robustness. As sensing corruption increased, obstacle avoidance decreased, participants passed closer to obstacles, workload increased, and ratings of confidence and feedback consistency declined. Qualitative results further suggest that the main issue was not only localization error itself, but the reduced interpretability of traversable free space, especially near boundaries and around dynamic obstacles. Taken together, these findings suggest that the practical success of SnowEcho in real-world deployment will depend less on perfect detection accuracy than on maintaining temporally stable and interpretable cues under noisy outdoor conditions.

6.4 Limitations

6.4.1 Feedback of SnowEcho. SnowEcho improved obstacle avoidance in controlled dynamic environments, but it has several limitations. First, while it performed well under the C_{Sparse} and C_{Dense} conditions, it was not equally effective under C_{Real}. One possible reason is that C_{Real} involved more complex and rapidly changing situations, including mixed static and dynamic obstacles as well as overtaking events. In such situations, users may need to understand the relative layout of multiple nearby hazards at once. However, our feedback design intentionally avoided presenting multiple simultaneous object sounds, because our pilot study suggested that such cues are difficult to process. One possible direction for improvement is therefore not simply to play multiple object sounds simultaneously, but to use different sound types or a more aggregated cue design to indicate that multiple nearby obstacles are present.

Moreover, participants reported cases in which it was difficult to understand the traversable space. When an obstacle approaches while the user is skiing near the course boundary, the system does not explicitly indicate whether there is sufficient clearance to pass between the boundary and the obstacle, which may cause hesitation. This limitation becomes more critical in situations closer to C_{Real}, where safe corridors change frequently depending on obstacle motion and local density.

In addition, the mapping between sound and distance/risk is not always intuitive and may require training. Because the perceived correspondence between sound and distance varies across individuals, the feedback may need to be personalized based on each user's skiing speed and preferences. One possible improvement is to adjust the urgency of the feedback not only based on obstacle distance but also on the estimated time remaining before reaching the obstacle. In skiing, the same distance can imply different levels of danger depending on speed, so a time-to-contact-based mapping may better reflect perceived risk.

6.4.2 Ski Simulator Environment. In our experiment, we validated the effectiveness of SnowEcho using a ski simulator. However, the simulator environment imposes constraints that differ from real-world skiing. Below, we summarize the main limitations.

First, the travel speed was fixed, and participants could not actively control their speed. In an actual ski resort, when a skier perceives danger but cannot immediately decide an avoidance direction, they can temporarily reduce risk by slowing down or stopping. In contrast, in our simulator, participants could not stop even if they wanted to, which effectively forced avoidance behavior to be expressed as a left–right decision.

Second, due to the simulator's physical width of 6 m, we could not set the course width in the VR environment to be sufficiently large. Although we set the VR course width to 12 m, participants approached the course boundaries relatively frequently, causing Side Audio to be presented earlier and more often. In real ski resorts, where slopes are typically wider, situations in which "Side Audio is heard immediately" are less likely to occur.

6.5 Future Work

As future work, we plan to expand SnowEcho in three directions. First, we will personalize the audio mapping to individual skiers by adapting parameters such as the distance and switching thresholds of Object Audio and Side Audio, and the degree of temporal smoothing, according to users' speed, experience, and auditory preferences. Second, we will refine the feedback based on participant comments, particularly by making traversable free space easier to understand and by adding higher-level contextual cues such as course progress, slope changes, and self-motion information. We will also revisit the design of rear-passing alerts and make them adaptive or optional when appropriate. Finally, we will move beyond simulator-based evaluation and validate the full system on real ski slopes to assess its safely, robustness, and usability under real-world sensing and skiing conditions.

7 Conclusion

The goal of this research is to move beyond the assumption that a sighted guide is required and enable people with visual impairments to ski independently and safely. In environments with dense dynamic obstacles, we showed that SnowEcho, which conveys obstacle locations via spatial audio, can be more effective than conventional route-guidance methods. Our findings further suggest that supporting active decision-making is better achieved by making the surroundings perceptible, rather than providing explicit directions. We also showed that this interaction remains effective under *Moderate Noise*. These results indicate that spatial-audio-based scene awareness is a promising approach for safer and more autonomous blind skiing, provided that cue stability is maintained under sensing uncertainty. Ultimately, we aim to contribute to inclusive ski assistance that allows anyone to challenge themselves on snow with greater autonomy.

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